

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Dept: CSE - AIDS | Course: DSA06 - Data Handling and Visualization

Set 1: Monthly Sales Data

Set 1: Monthly Sales Data

Dataset: Month vs Sales (in \$) — January 15000, February 18000, March 22000, April 20000, May 23000.

Question 1

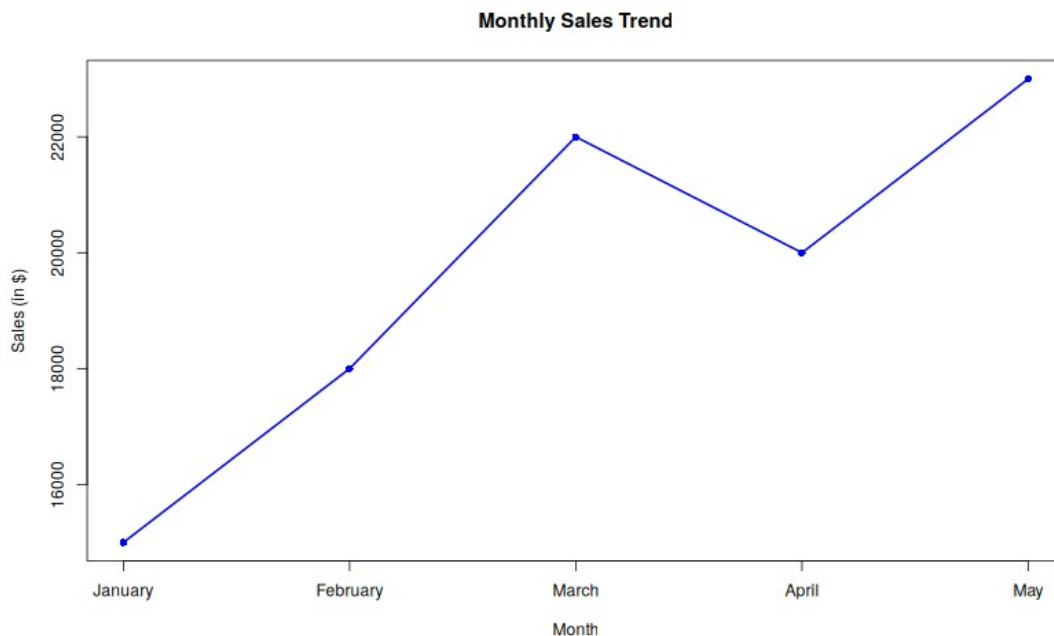
Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

Code

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(1:5, sales, type="o", col="blue", lwd=2, pch=16, xaxt="n",
     xlab="Month", ylab="Sales (in $)", main="Monthly Sales Trend")
axis(1, at=1:5, labels=month)
```

Output



Question 2

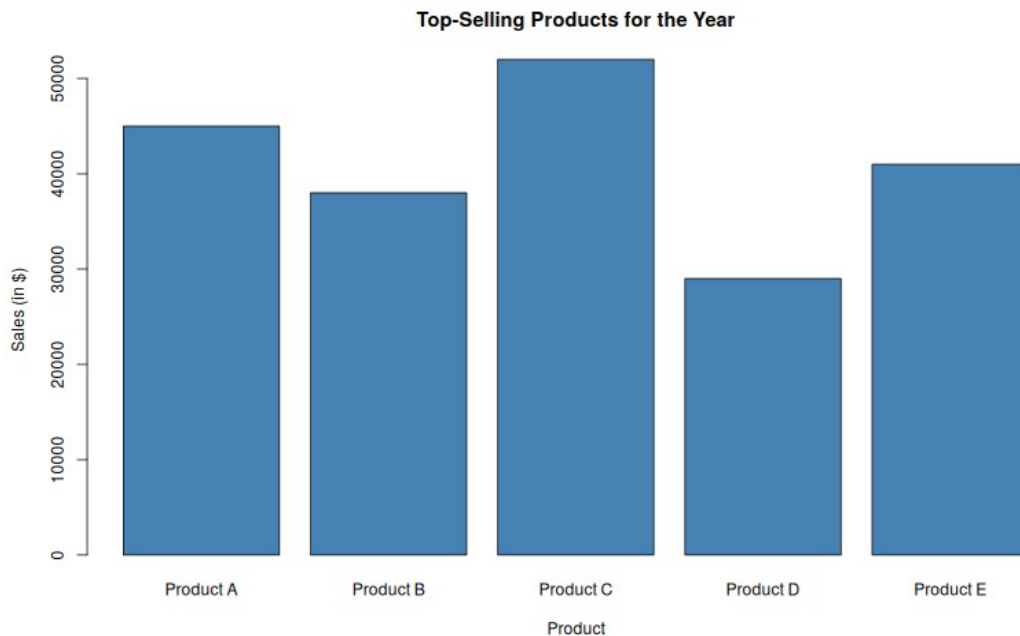
Generate a bar chart to display the top-selling products for the year. Label the chart elements.

Code

```
products <- c("Product A","Product B","Product C","Product D","Product E")
product_sales <- c(45000, 38000, 52000, 29000, 41000)

barplot(product_sales, names.arg=products, col="steelblue",
        main="Top-Selling Products for the Year",
        xlab="Product", ylab="Sales (in $)")
```

Output



Note: product-level sales were not given in the dataset, so representative sample figures were used.

Question 3

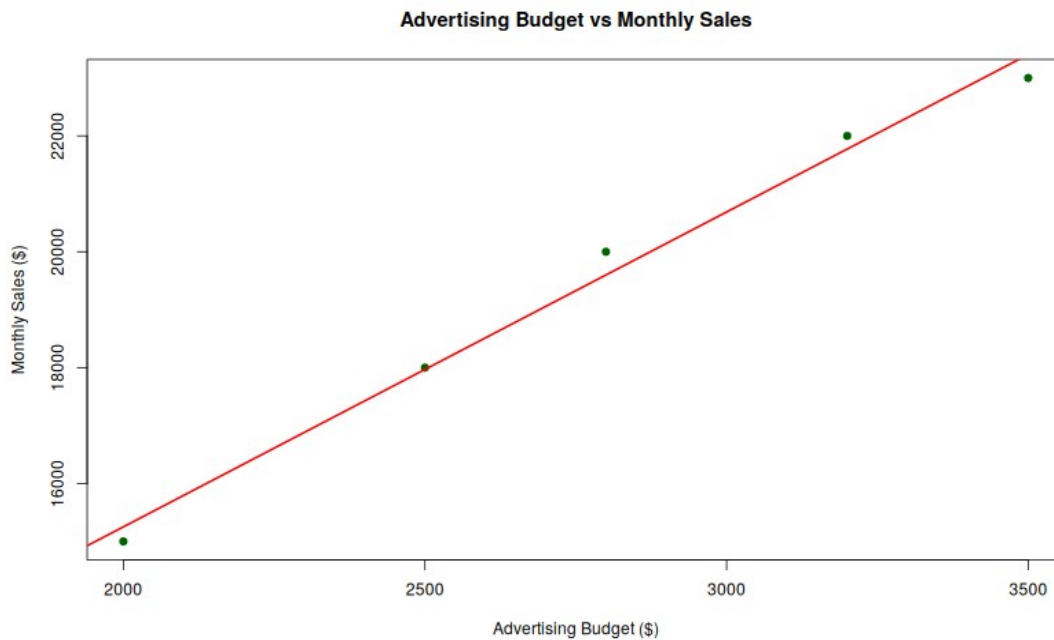
Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

Code

```
ad_budget <- c(2000, 2500, 3200, 2800, 3500)

plot(ad_budget, sales, pch=19, col="darkgreen",
     xlab="Advertising Budget ($)", ylab="Monthly Sales ($)",
     main="Advertising Budget vs Monthly Sales")
abline(lm(sales ~ ad_budget), col="red", lwd=2)
```

Output



Insight: The upward-sloping trend line shows a positive relationship — as advertising budget increases, monthly sales tend to increase as well.

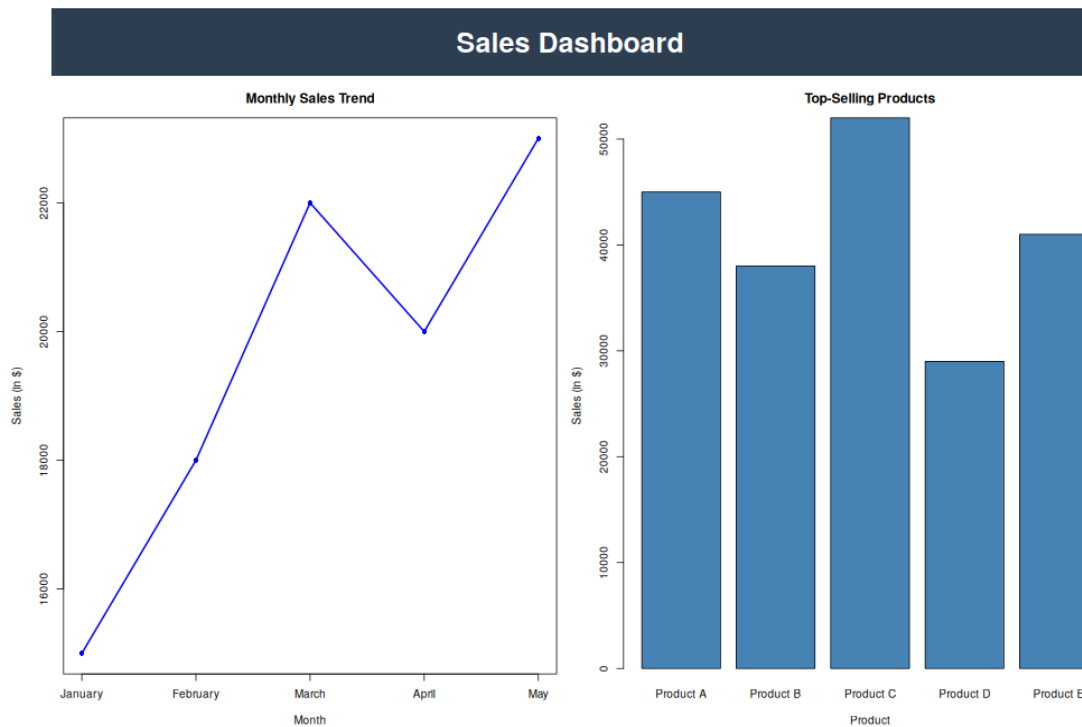
Question 4

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

Code

```
library(shiny)
ui <- fluidPage(
  titlePanel("Sales Dashboard"),
  plotOutput("lineChart"),
  plotOutput("barChart")
)
server <- function(input, output) {
  output$lineChart <- renderPlot({
    plot(1:5, sales, type="o", col="blue", xaxt="n",
         xlab="Month", ylab="Sales", main="Monthly Sales Trend")
    axis(1, at=1:5, labels=month)
  })
  output$barChart <- renderPlot({
    barplot(product_sales, names.arg=products, col="steelblue",
            main="Top-Selling Products")
  })
}
shinyApp(ui, server)
```

Output



Running this Shiny app opens a live browser dashboard; the rendered view above shows both charts combined under one dashboard panel as the app displays them.